SOLAX POWER

Connection of Datahub

ENGLISH

V1.0

Datahub

DataHub, a special equipment of the monitoring platform of photovoltaic power generation system, has realized many functions, with details as follows: interface aggregation, data acquisition, data storage, output control, and centralized monitoring and centralized maintenance of inverters, electricity meters, environmental monitors and other equipment in photovoltaic power generation systems.The specific interface definition of Datahub is shown in the following figure.

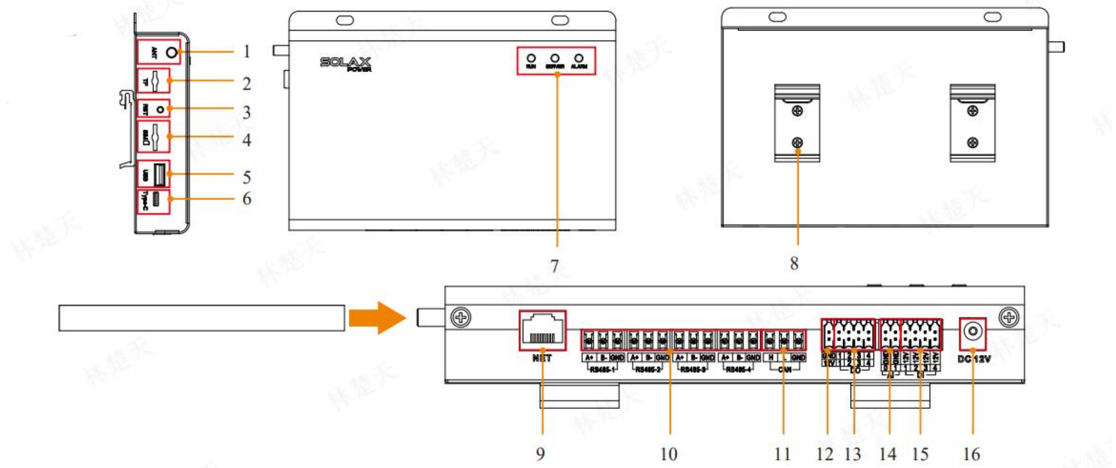


Figure 1 Datahub Interface

Table 1 Datahub Interface

Interface	Interface Definition
1	Antenna jack
2	TF card socket (TF)
3	RST button (RST)
4	SIM card socket (SIM)
5	USB socket (USB)
6	TYPE-C socket
7	LED indicator (RUN,SERVER,ALARM)
8	Rail clip

9	NET socket (NET)
10	RS485 socket (RS485)
11	CAN socket (CAN)
12	12V power output (12V/GND)
13	DO socket (DO)
14	AI socket (AI)
15	DI socket (DI)
16	12V power input (DC12V)

Datahub+EVC

The connection between the EV Charger and Datahub can realize the using Datahub as device unification to achieve the charging current control of multiple EV Charger.As shown in the figure below,where the blue wire is the communication wire and the red wire is the AC wire.Firstly,the INV/CT port of the EV Charger is connected to one RS485 port of the Datahub,and the communication port RS485 of the meter is connected to one RS485 port of the Datahub.Next,connect the EV Charger and meter to the AC side.

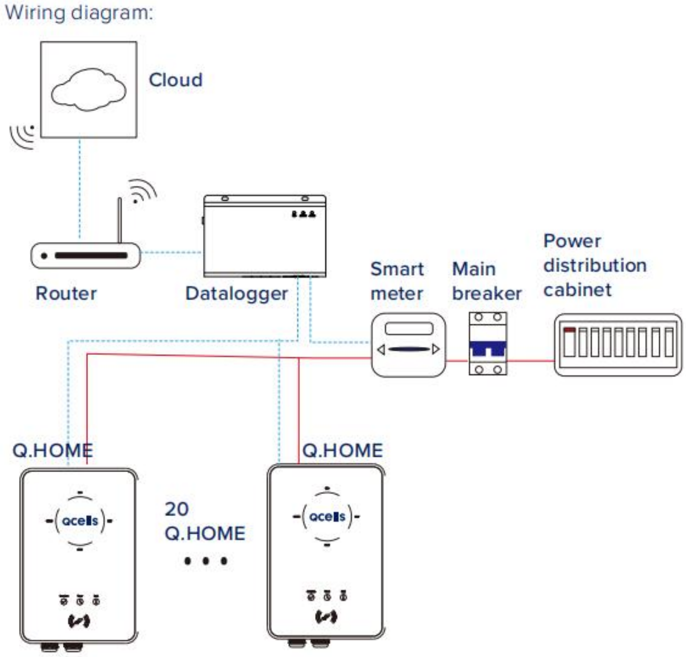


Figure 1 Datahub+EVC Scheme

Datahub+EVC+On-grid Inverter+Hybrid

The scheme of connecting EV Charger,Datahub,on-grid inverter and energy storage inverter is another method of using Datahub to achieve unified management of multiple devices.As shown in the figure below,X3-Forth and X3-Hybrid G4 are used as an example.The first is the connection between the on-grid inverter and Datahub.The PIN4,5 of the first inverter(the one closest to Datahub) communication port RS485-1 is connected to the PIN1,2 of the second inverter communication port RS485-1,and the PIN4,5 is connected to the PIN1,2 of the next inverter,and so on.PIN1,2 of the first inverter communication port RS485-1 is connected to any RS485 port of Datahub.The meter is connected to any RS485 port of the Datahub.Next is the connection between Hybrid and Datahub.The CAN2 port of the master hybrid connects to the CAN1 port of the next slave,and so on.The CAN1 port of the master hybrid connects to the EPS Parallel Box(if there is an EPS load).Meter(address 1)is connected to Meter/CT port of master hybrid.COM port PIN4,5 of master hybrid is connected to any RS485 port of Datahub(the red solid line represents the AC line, the blue dashed line represents the communication line, and the green dashed line represents the CAN line).

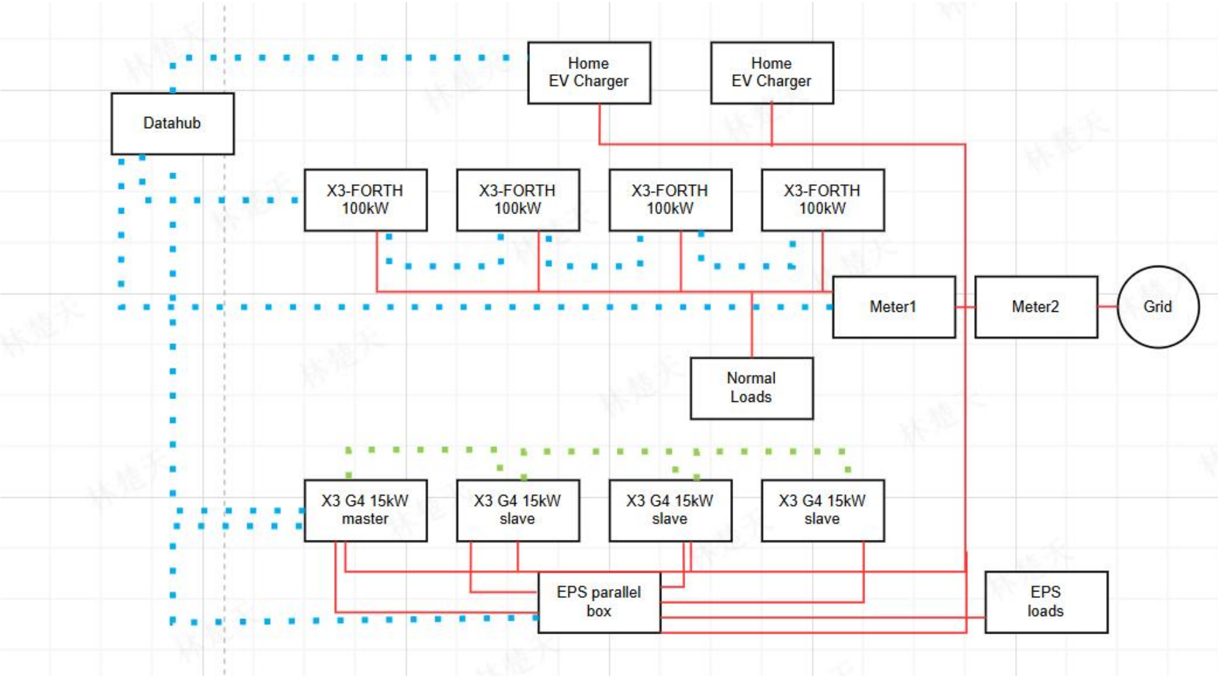


Figure 2 Datahub+EVC+On-grid Inverter+Hybrid

Datahub+EVC+Adapter Box G2+Hybrid

The scheme of connecting EV Charger,Datahub,Adapter Box G2 and energy storage inverter is the third method of using Datahub to achieve unified management of multiple devices.The X3-Hybrid G4 is used as an example for the energy storage inverter.The connection of the energy storage inverter as well as the EV Charger and Datahub is the same as the above scheme.The connection between the Adapter Box G2 and Datahub is through the 485_INV port of the Adapter Box G2 and any RS485 port of Datahub.The figure of this scheme is showing as follows(The Adapter Box G2 and Datahub currently only support use in smart scene)(the red solid line represents the AC line, the blue dashed line represents the communication line, and the red dashed line represents the control line).

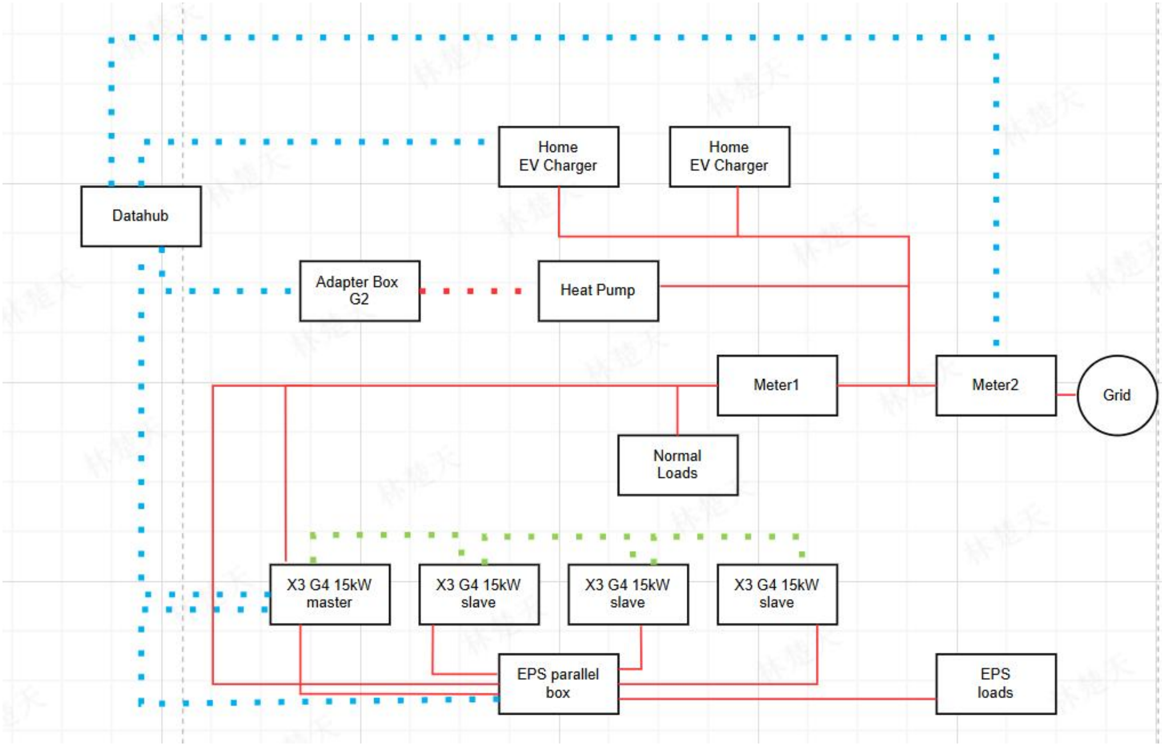


Figure 3 Datahub+EVC+Adapter Box G2+Hybrid

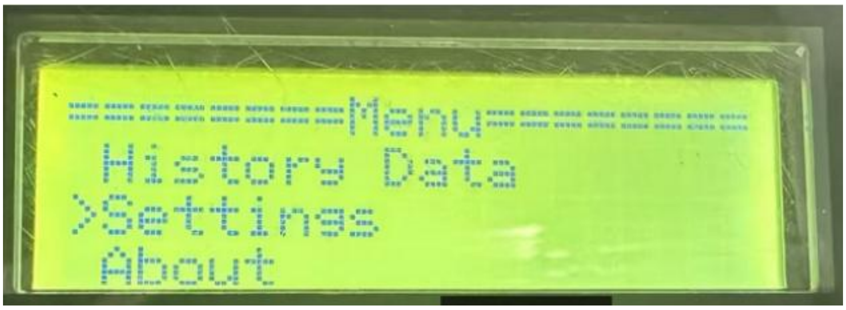
Specific Setup

The COM port of the master X3G4 communicates with the Datahub485(crystal head wiring sequence is from blue-white/blue, DataHub port wiring is blue-white/blue).The remaining slaves are connected to the master with 485 (CAN port), make sure X3G4s have the baud rate 19200, the master address is set to 1, and the remaining slaves have the address set from 2.

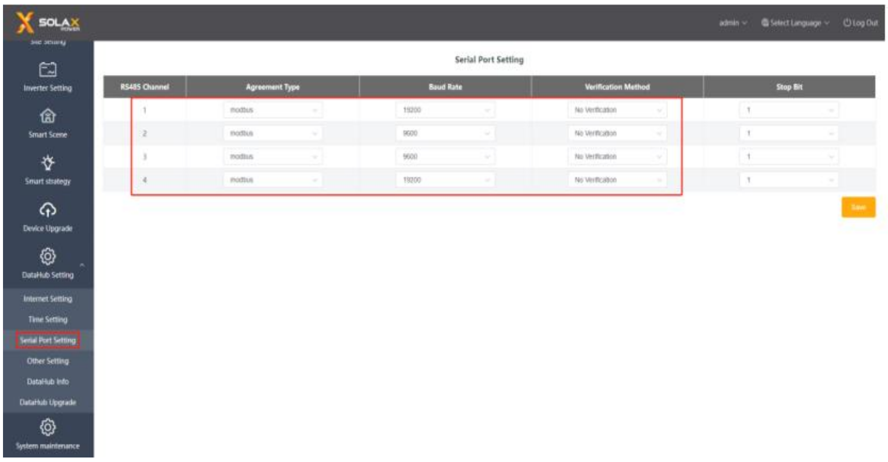
DataHub is connected to 485in of the first forth (no crystal head required), 485out is connected to 485in of the second forth, 485out is connected to 485in of the third forth, and so on until the forth is connected; make sure that the baud rate of the forth and the DataHub is 9600.The forth device can be added automatically by assigning the address in the station management - add device, so there is no need to change the address of the forth device manually.

DataHub and EVC are connected (crystal head is needed), connect a 485 communication cable from each COM port of 20 EVCs to DataHub485, make sure that the baud rate of EVCs and DataHub is 9600.EVCs will be added with an address automatically in the station management - add device, so you don't need to change the address of the EVCs manually.

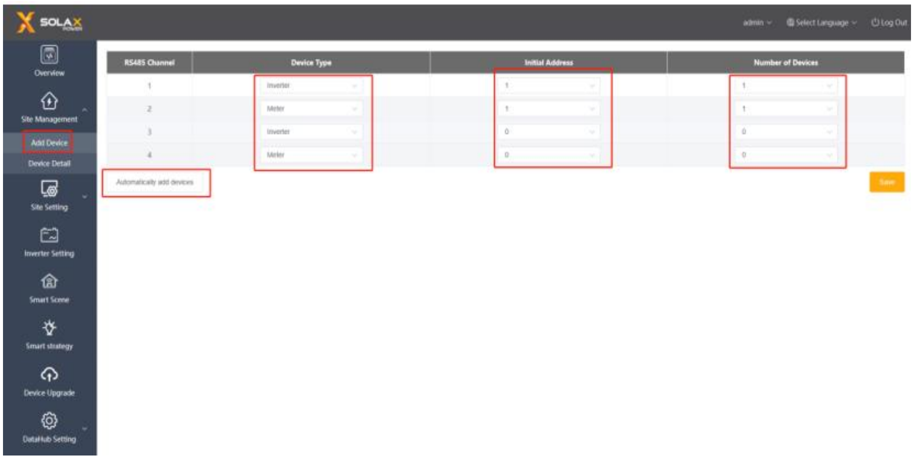
X3G4 Procedure for Setting Baud Rate and Address



DataHub Setting Baud Rate and Adding Devices



Note: Enter the local interface to select - System Setting - Serial Port Setting to confirm the baud rate of the device.



Note:After confirming, enter - Site Management - Add Device to select the device connected under the corresponding serial port, confirm that the starting address is 1, and the quantity is the total number of the connected device. (EV Charger and FORTH machine can use the function of automatic address allocation but you need to confirm whether the equipment and quantity are correct).